

Standard Form for Per-Unit-Length RLGC Matrices of Multiconductor Transmission Lines

Theory, Matrix Structure, and Physical Interpretation

Abstract

This document presents the standard per-unit-length (PUL) matrix formulation for multiconductor transmission lines (MTLs), following Clayton R. Paul's *Analysis of Multiconductor Transmission Lines* [1]. For a uniform line with n signal conductors and a reference conductor (conductor 0), the coupled Telegrapher equations describe voltage and current along the line under the transverse electromagnetic (TEM) or quasi-TEM (mode) approximation. The model is specified by four $n \times n$ matrices: inductance $\mathbf{L}$, resistance $\mathbf{R}$, conductance $\mathbf{G}$, and capacitance $\mathbf{C}$. This document gives the governing equations, physical interpretations, and standard matrix forms.

1. Multiconductor Transmission Line (MTL) System

Consider a uniform multiconductor transmission line consisting of n signal conductors and an additional conductor, designated conductor 0 (i.e., the reference conductor or ground conductor), that serves as the common reference (return) conductor [1]. The line geometry and surrounding medium are assumed uniform along the longitudinal propagation axis z .

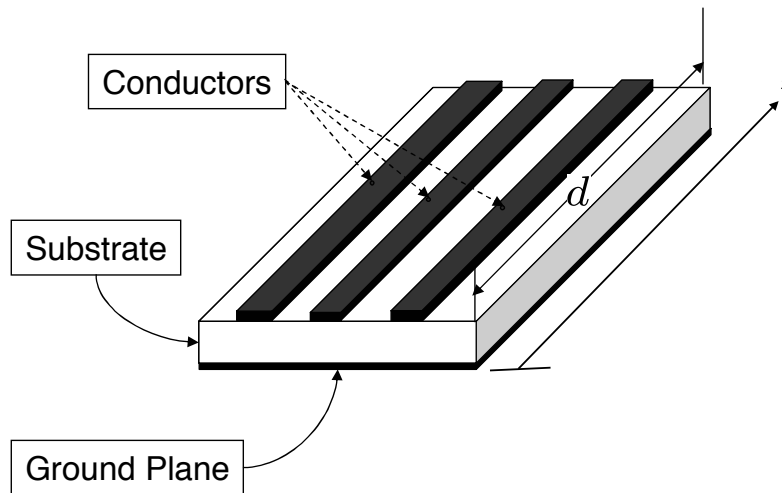


Figure 1: Geometry of a uniform multiconductor transmission line with n signal conductors and a reference conductor (conductor 0).

The line voltages of the n signal conductors with respect to the reference conductor are represented by the $n \times 1$ voltage vector $\mathbf{V}(z, t)$, and the signal conductor currents are represented by the $n \times 1$ current vector $\mathbf{I}(z, t)$. These are defined in (1) as

$$\mathbf{V}(z, t) = \begin{bmatrix} v_1(z, t) \\ v_2(z, t) \\ \vdots \\ v_n(z, t) \end{bmatrix}, \quad \mathbf{I}(z, t) = \begin{bmatrix} i_1(z, t) \\ i_2(z, t) \\ \vdots \\ i_n(z, t) \end{bmatrix}. \quad (1)$$

The total return current flowing through the reference conductor is given by Kirchhoff's Current Law:

$$i_0(z, t) = - \sum_{i=1}^n i_i(z, t).$$

All four matrices are per-unit-length quantities. The time-domain equations assume frequency-independent parameters; practical conductor and dielectric models may make the effective parameters frequency dependent.

2. Coupled Multiconductor Telegrapher Equations

The spatial distribution and temporal dynamics of the voltage and current vectors along the multiconductor line are governed by the coupled Telegrapher equations in matrix differential form [1]:

$$\frac{\partial \mathbf{V}(z, t)}{\partial z} = -\mathbf{R}\mathbf{I}(z, t) - \mathbf{L} \frac{\partial \mathbf{I}(z, t)}{\partial t}, \quad (2a)$$

$$\frac{\partial \mathbf{I}(z, t)}{\partial z} = -\mathbf{G}\mathbf{V}(z, t) - \mathbf{C} \frac{\partial \mathbf{V}(z, t)}{\partial t}. \quad (2b)$$

Using the $e^{j\omega t}$ phasor convention, with angular frequency ω and phasors $\hat{\mathbf{V}}(z)$ and $\hat{\mathbf{I}}(z)$, the equations become:

$$\frac{d\hat{\mathbf{V}}(z)}{dz} = -\mathbf{Z}\hat{\mathbf{I}}(z) = -(\mathbf{R} + j\omega\mathbf{L})\hat{\mathbf{I}}(z), \quad (3a)$$

$$\frac{d\hat{\mathbf{I}}(z)}{dz} = -\mathbf{Y}\hat{\mathbf{V}}(z) = -(\mathbf{G} + j\omega\mathbf{C})\hat{\mathbf{V}}(z). \quad (3b)$$

Here, $\mathbf{Z} = \mathbf{R} + j\omega\mathbf{L}$ is the $n \times n$ per-unit-length series-impedance matrix, and $\mathbf{Y} = \mathbf{G} + j\omega\mathbf{C}$ is the $n \times n$ per-unit-length shunt-admittance matrix. The PUL units are Ω/m for $\mathbf{R}$, H/m for $\mathbf{L}$, S/m for $\mathbf{G}$, and F/m for $\mathbf{C}$.

3. Longitudinal Parameter Matrices ($\mathbf{L}$ and $\mathbf{R}$)

The longitudinal parameters model the magnetic flux linkages and ohmic losses associated with current loops formed by the signal conductors and the reference conductor:

- **Per-Unit-Length Inductance Matrix, $\mathbf{L}$** ($n \times n$, symmetric positive definite for independent circuits):
The inductance matrix contains the individual per-unit-length self-inductances, l_{ii} , of the circuits formed

between conductor i and reference conductor 0 (i.e., ground), and the per-unit-length mutual inductances, $l_{ij} = l_{ji}$, between circuits i and j , as

$$\mathbf{L} = \begin{bmatrix} l_{11} & l_{12} & \cdots & l_{1n} \\ l_{21} & l_{22} & \cdots & l_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ l_{n1} & l_{n2} & \cdots & l_{nn} \end{bmatrix}. \quad (4)$$

- **Per-Unit-Length Resistance Matrix, $\mathbf{R}$** ($n \times n$, symmetric positive definite when all signal-conductor resistances are positive): The resistance matrix accounts for the finite conductivity of the conductors. The diagonal entries ($r_i + r_0$) represent the total round-trip loop resistance for circuit i consisting of signal conductor resistance r_i and the common return conductor resistance r_0 . The off-diagonal entries r_0 represent the common-impedance resistive coupling due to shared return current in conductor 0:

$$\mathbf{R} = \begin{bmatrix} r_1 + r_0 & r_0 & \cdots & r_0 \\ r_0 & r_2 + r_0 & \cdots & r_0 \\ \vdots & \vdots & \ddots & \vdots \\ r_0 & r_0 & \cdots & r_n + r_0 \end{bmatrix}. \quad (5)$$

where $\mathbf{V}$ and $\mathbf{I}$ are defined by (1).

4. Transverse Parameter Matrices ($\mathbf{G}$ and $\mathbf{C}$)

The transverse parameters represent the conduction and displacement currents flowing across the dielectric medium in the transverse cross-sectional plane:

- **Per-Unit-Length Conductance Matrix, $\mathbf{G}$** ($n \times n$, symmetric): Represents transverse conduction current (dielectric leakage losses) flowing between the conductors in the transverse plane as

$$\mathbf{G} = \begin{bmatrix} \sum_{k=1}^n g_{1k} & -g_{12} & \cdots & -g_{1n} \\ -g_{21} & \sum_{k=1}^n g_{2k} & \cdots & -g_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -g_{n1} & -g_{n2} & \cdots & \sum_{k=1}^n g_{nk} \end{bmatrix}. \quad (1)$$

Let $g_{ij} = g_{ji} \geq 0$ be the conductance between signal conductors i and j , and let $g_{i0} \geq 0$ be the conductance between signal conductor i and the reference. Then

$$\mathbf{G} = [G_{ij}], \quad G_{ij} = \begin{cases} g_{i0} + \sum_{\substack{k=1 \\ k \neq i}}^n g_{ik}, & i = j, \\ -g_{ij}, & i \neq j. \end{cases} \quad (6)$$

- **Per-Unit-Length Capacitance Matrix, $\mathbf{C}$** ($n \times n$, symmetric positive definite for a well-posed cross-section referenced to conductor 0): Represents transverse displacement current (electric-field coupling) flowing between the conductors in the transverse plane as

$$\mathbf{C} = \begin{bmatrix} \sum_{k=1}^n c_{1k} & -c_{12} & \cdots & -c_{1n} \\ -c_{21} & \sum_{k=1}^n c_{2k} & \cdots & -c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -c_{n1} & -c_{n2} & \cdots & \sum_{k=1}^n c_{nk} \end{bmatrix}. \quad (2)$$

Let $c_{ij} = c_{ji} \geq 0$ be the partial capacitance between signal conductors i and j , and let $c_{i0} \geq 0$ be the partial capacitance between signal conductor i and the reference. Then

$$\mathbf{C} = [C_{ij}], \quad C_{ij} = \begin{cases} c_{i0} + \sum_{\substack{k=1 \\ k \neq i}}^n c_{ik}, & i = j, \\ -c_{ij}, & i \neq j. \end{cases} \quad (7)$$

Matrix structure. For both transverse matrices, each diagonal entry is the sum of all branches incident on that conductor, including its branch to conductor 0; each off-diagonal entry is the negative of the mutual branch parameter. Thus the matrices have nonpositive off-diagonal entries and are diagonally dominant. The conductance matrix is positive semidefinite, and is positive definite when every connected component of the transverse conductance network has a path to the reference. The capacitance matrix is positive definite for a well-posed cross-section referenced to conductor 0.

References

- [1] C. R. Paul, *Analysis of Multiconductor Transmission Lines*. New York, NY, USA: John Wiley & Sons, Inc., 1994, ch. 2, pp. 51–55.